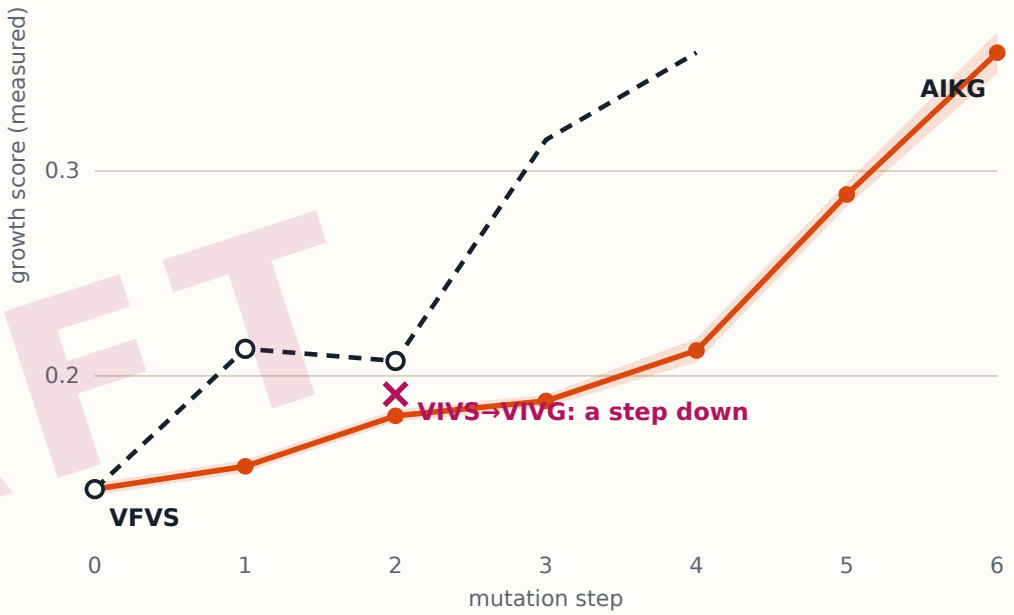
DRAFT — replay not validated

A protein design is not a plan.

Which mutation do you make first — and what evidence says the next step will still work?

TARGET AND THE MISSING ROUTE

TrpB enzyme, four residues, parent VFVS → best measured variant AIKG. All four changes are needed, so there are **24** orders to make them in. Taking the measurements at face value, **every one** of the 24 steps down at least once.



--- the most tempting direct ordering VFVS → VIVS → VIVG → VIKG → AIKG

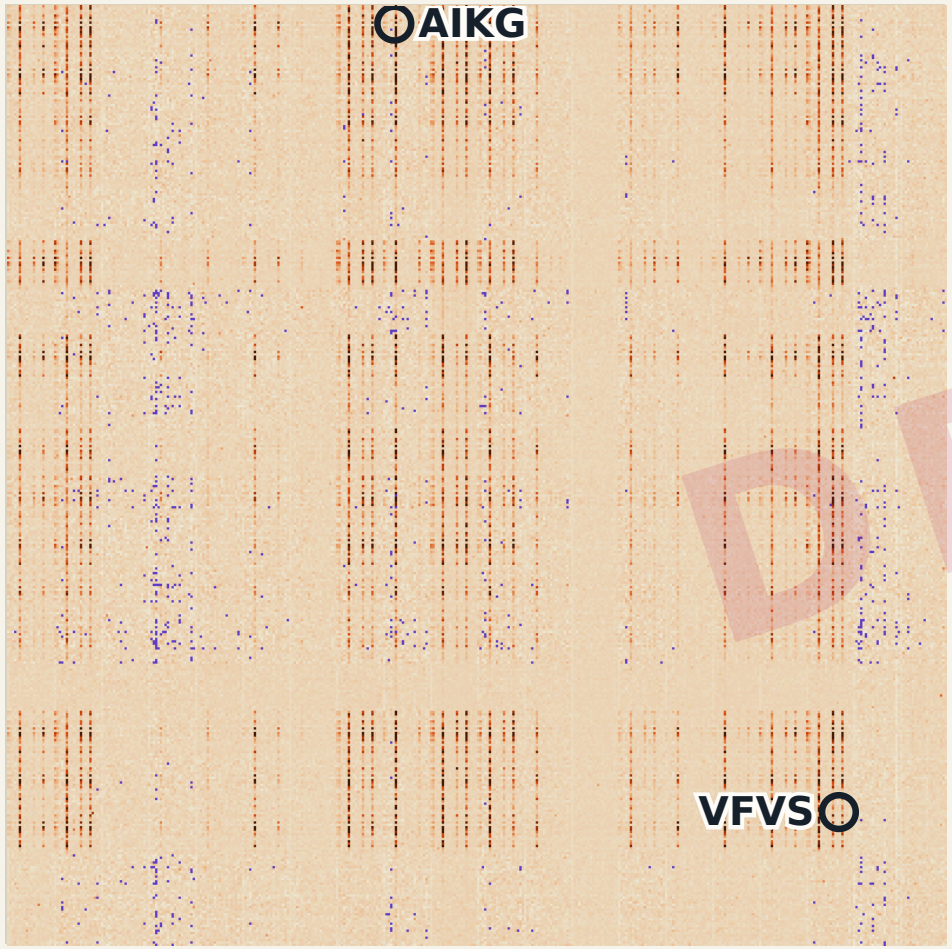
— the 6-step detour whose every step is certified (band = assumed κ=2 interval)

DRAFT

Source Johnston et al., PNAS 2024, doi 10.1073/pnas.2400439121; data CaltechDATA 10.22002/h5rah-5z170 (CC0), dataset summary_AA.csv sha256 a631d8cf6c5b… · orderings recounted from site/data/landscape.js, cross-checked with artifacts/evidence/route_summary.json · artifacts @ 4398fad0

DRAFT — replay not validated

This is a measured landscape: 159,129 of 160,000 variants have a laboratory value.



159,129
measured genotypes, two replicate flasks each

871
never measured — stay unknown

ROW LEDGER	COUNT
rows in source file (incl. stop codons)	193,170
stop-codon rows, excluded	34,041
measured, no stop	159,129
imputed in the authors' separate file	871
imputed rows used as observations here	**0**

Colour = measured growth score (cream low → deep orange high). The 871 unmeasured cells are violet and too small to see: each is a place a route could pass that we have no data for.

Reads as: cellular growth of E. coli selecting for tryptophan synthase activity — a proxy, not a purified-enzyme constant. Layout: rows = residues 1×2, columns = residues 3×4. A projection of sequence space, not molecular geometry.

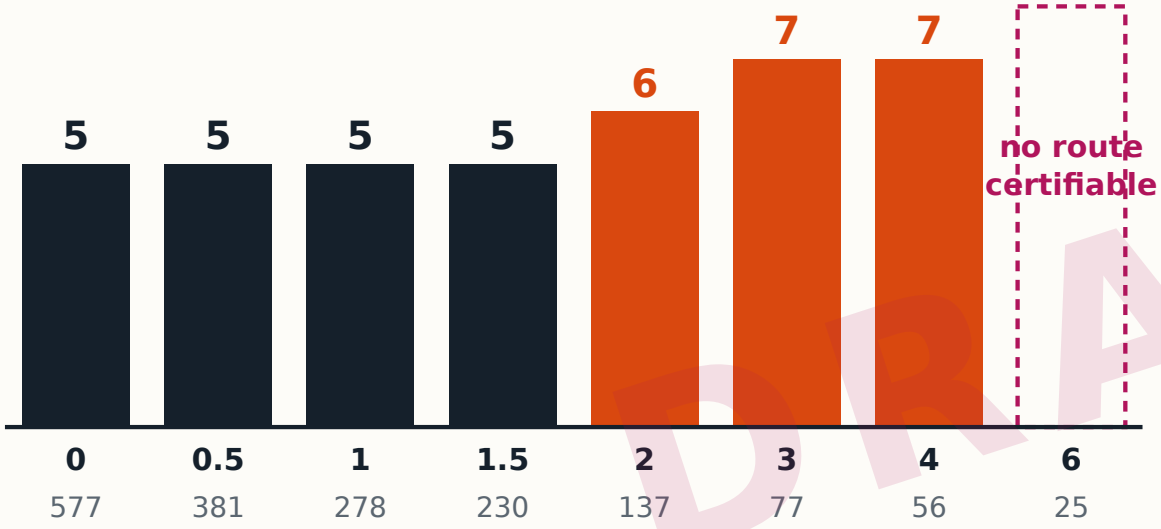
DRAFT

Source dataset summary_AA.csv sha256 a631d8cf6c5b…; counts from artifacts/demo.json source.counts (independent recount: scripts/audit_data.py) · score = mean of two replicates' growth rate minus stop-codon background; not max-normalised · artifacts @ 4398fad0

DRAFT — replay not validated

Challenge the assumption and the route moves: 5 steps → 6 → 7, then no route can be certified.

mutations on the shortest route whose every step is certified



bold row: assumed κ · grey row: variants reachable from VFVS by certified steps

κ multiplies the replicate-disagreement spread to widen every measurement into an interval. It is an **assumed** sensitivity — not a confidence level. A step is certified only if the next variant's lower bound clears the current one's upper bound.

The weakest step of the κ=2 route

SFMS → SIMS clears the bar by only **0.0013** growth units; the route is certifiable up to κ≈**2.45** and must be re-planned beyond.

Click a step in the live tool and the raw replicate rows and read counts for that variant appear.

Precisely: at κ=6 the tool says "no route certifiable at this bound" (25 variants reachable by certified steps). That does *not* mean no ascending route exists: overlapping intervals are simply not certified. κ=4 is the last bound in the sweep with a certified route.

When the evidence cannot settle a step, the system's output is the next assay to run — not a confidence number it cannot support.

DRAFT

Source artifacts/demo.json kappa_sweep (core worker; frozen noise model artifacts/core/protocol.json), independently recomputed at all 8 κ by scripts/independent_truth.py · graph scope: measured 4-site subgraph, 871 unmeasured genotypes outside · artifacts @ 4398fad0

DRAFT — replay not validated

DRAFT — matched-budget replay result pending the integrated 20-seed run.

REPLAY RESULT NOT FINAL

No replay numbers are shown: the integrated 20-seed run failed validation. This slot refreshes automatically when `build_deck.py` passes.

DRAFT

- summary.json seeds=5 != protocol seeds=20
- 105 of 140 replay shards have zero rows (e.g. ['collective_isolated__seed05.json', 'collective_isolated__seed06.json', 'collective_isolated__seed07.json'])
- random: seeds present [0, 1, 2, 3, 4] != 0..19
- random: 25 episodes != 100
- greedy: seeds present [0, 1, 2, 3, 4] != 0..19

DRAFT — replay not validated

The 6-step route is checked by the Lean kernel — conditional on the intervals we assumed.

WHAT THE THEOREM SAYS

If every true value lies inside its interval, and each step satisfies `lower(next) ≥ upper(current)`, then the route never goes down.

A second theorem: a closed set whose every exit is certified lower rules out any non-decreasing path. Both are generic over any graph.

RECEIPT	RESULT
THIS ROUTE — core κ=2 bridge receipt	
original route builds in Lean	`exit 0`
swapped-route mutant	`rejected, exit 1`
intervals recomputed from raw CSV	7 / 7
THE THEOREMS — proof receipt	
kernel compile of the original	`exit 0`
dominance-flip and open-cut mutants	`rejected` exits 1, 1
instance mutants as expected · source mutants rejected	35/35 · 8/8
axioms used	`Quot.sound, propext`

Route VFVS → AIKG at assumed κ=2

STEP	MARGIN	READS
VFVS → VFMS	0.0052	261
VFMS → SFMS	0.0186	265
SFMS → SIMS	0.0013	248
SIMS → SIMG	0.0157	178
SIMG → SIKG	0.0644	135
SIKG → AIKG	0.0534	27

margin = lower(next) − upper(current), growth units; reads = lower of the two replicates' input counts.

Covered: this route's seven intervals, in Lean. **Assumed:** that those intervals contain the truth; two replicates give no confidence bands. **Not covered:** the replay's decisions — a replay "blocked" means "nothing certifiable", not the Lean no-path theorem. Four sites only; nothing here says anything about sequence space.

DRAFT

Source artifacts/proof/receipt.json (Lean 4.35.0, check_proofs wall 192 s) and artifacts/integration/core_kappa2_route_lean.json · a static receipt shows a compiled check, not a proof that the live slider is right · artifacts @ 4398fad0

DRAFT — replay not validated

The wedge: a decision layer that says which assay, and why it is justified — tested next on a real enzyme campaign.

BUYER (HYPOTHESIS)

Economic buyer: head of protein engineering at an industrial-enzyme or biocatalysis group, or a CRO running variant campaigns.

Pays for: fewer wasted assays and an auditable reason for each one, above the protein models they already use.

Today: a retrospective tool on one public landscape. **Future, unproven:** a wet-lab cost or time advantage.

NEXT REAL EXPERIMENT: ONE-LAB PROSPECTIVE PILOT

Lab supplies a prior campaign and a new target; we pre-register the decision rule, then both arms spend the same assay budget.

DAYS	MILESTONE
0–30	data agreement; pre-registered protocol and endpoint
31–60	blind retrospective replay on the lab's own history
61–100	two prospective rounds: tool-guided vs lab-usual
101–120	readout: cost and days to a defensible decision

THE ASK

One enzyme lab. One prior campaign. One prospective round.

No customer, LOI or market size exists yet; the moat, if any, is prospective evidence and workflow adoption, not Lean alone.

DRAFT

Biologic design starts with justified enzyme experiments — one measured decision at a time, not text-to-organism.